Comprehension Quizzes with Answer Options

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This PDF documents the comprehension quizzes integrated into the application. Each item includes all answer options and the correct answer used for scoring.

Article 1: Webb's Scientific Instruments

1. Where are Webb's scientific instruments located?

A. Behind the primary mirror in a separate external unit

B. Within the Integrated Science Instrument Module.

C. Inside the sunshield layers that block heat from the Sun

D. On a ground station that receives Webb's observations

Correct answer: Within the Integrated Science Instrument Module.

2. Why are Webb's scientific instruments important?

A. They keep the telescope pointed only toward nearby planets

B. They replace the telescope mirrors during long observations

C. They record and analyze light for scientific observations.

D. They store fuel and control Webb's launch sequence

Correct answer: They record and analyze light for scientific observations.

3. Which of the following is one of Webb's four main scientific instruments?

A. NIRCam.

B. SolarNet Imaging Module

C. Deep Field Navigation Camera

D. Guided Planet Mapper

Correct answer: NIRCam.

4. What kind of light is Webb able to observe?

A. Mostly radio waves from nearby satellites

B. Only visible light from planets and stars

C. Mainly ultraviolet light from hot stars

D. Infrared light.

Correct answer: Infrared light.

5. What happens to light during an observation with Webb?

A. It is stored in the sunshield before being sent to Earth

B. It is gathered by the mirrors and analyzed with scientific instruments.

C. It is converted into sound before any measurement is made

D. It is reflected away before reaching the instrument module

Correct answer: It is gathered by the mirrors and analyzed with scientific instruments.

6. Why does the author describe Webb's instruments as Swiss army knife?

A. Because the instruments are used mainly for physical repairs

B. Because each instrument has only one simple observation task

C. Because each instrument has several specialized observing modes.

D. Because the telescope carries backup tools for emergencies

Correct answer: Because each instrument has several specialized observing modes.

7. What do cameras in Webb's instruments do?

A. They block bright stars before light reaches the mirrors

B. They convert every observation into a wavelength graph

C. They control the telescope's position during launch

D. Capture two-dimensional images of regions of space.

Correct answer: Capture two-dimensional images of regions of space.

8. What is a spectrograph?

A. Tool for spreading light into a spectrum in order to measure various wavelengths.

B. To keep the telescope cold by filtering heat from the Sun

C. To take standard two-dimensional pictures of space objects

D. To lock Webb onto guide stars during every observation

Correct answer: Tool for spreading light into a spectrum in order to measure various wavelengths.

9. What do coronagraphs allow Webb to observe?

A. The internal temperature of Webb's science module

B. The exact launch path used to reach space

C. Faint planets or debris disks around bright stars.

D. Only large galaxies that fill the full field of view

Correct answer: Faint planets or debris disks around bright stars.

10. What do detectors do in Webb's instruments?

A. They physically clean dust from the primary mirror

B. Absorb light and convert it into digital data.

C. They move the telescope between different orbits

D. They create artificial light for calibration images

Correct answer: Absorb light and convert it into digital data.

Article 2: A "Robot Pizza Chef" Serving Up Better Quantum Computers

1. What is the primary function of the QIS cluster tool at Berkeley Lab?

A. To automate food preparation for researchers in the lab

B. To assemble ordinary desktop computers at high speed

C. Help researchers develop quantum computer components.

D. To replace material experiments with software simulations only

Correct answer: Help researchers develop quantum computer components.

2. Where is the QIS cluster tool located?

A. At the Molecular Foundry.

B. Inside a commercial smartphone factory

C. On the International Space Station

D. At a public demonstration restaurant

Correct answer: At the Molecular Foundry.

3. Why are qubits hard to make?

A. They work best when touched and adjusted by hand

B. They are usually made from ordinary plastic parts

C. They do not require controlled materials or surfaces

D. They are extremely sensitive to the environment and any defects.

Correct answer: They are extremely sensitive to the environment and any defects.

4. What is the advantage of QIS cluster tool in that it combines fabrication and analysis in one vacuum system?

A. It removes the need to measure materials after fabrication

B. Reduces contamination and makes the process faster and more standard.

C. It allows quantum devices to be fabricated outdoors

D. It makes each qubit larger and easier to inspect visually

Correct answer: Reduces contamination and makes the process faster and more standard.

5. What does the robotic arm move in the QIS cluster tool?

A. An 8-inch wafer.

B. A finished laptop

C. A pizza tray

D. A telescope lens

Correct answer: An 8-inch wafer.

6. What is a Josephson junction?

A. A camera system used to guide a space telescope

B. A software tool used to organize experimental data

C. A standard battery component used in mobile phones

D. Very tiny element consisting of two superconductors with an ultrathin insulator between them.

Correct answer: Very tiny element consisting of two superconductors with an ultrathin insulator between them.

7. Why are Josephson junctions needed in quantum computing research?

A. They are used only to create images of material surfaces

B. They make classical computers run ordinary programs faster

C. To create circuits that could serve as qubits.

D. They replace the need for low-temperature experiments

Correct answer: To create circuits that could serve as qubits.

8. What is the role of artificial intelligence in this research?

A. It can fabricate qubits without any physical materials

B. Artificial intelligence can learn from fabrication data and make suggestions about materials and production techniques.

C. It can turn quantum computers into ordinary calculators

D. It can remove the need to run experiments or analyze samples

Correct answer: Artificial intelligence can learn from fabrication data and make suggestions about materials and production techniques.

9. What kinds of analytical tools are used in the QIS cluster tool?

A. Electrons, x-rays, lasers, and infrared light.

B. Only a standard phone camera and room lighting

C. Kitchen scales, thermometers, and pressure gauges

D. Only visual inspection by researchers through glass

Correct answer: Electrons, x-rays, lasers, and infrared light.

10. What else, aside from quantum computers, could these sensitive quantum devices be used for?

A. They could replace all internet cables in buildings

B. They could be used only for gaming graphics

C. As sensors.

D. They could function as ordinary clothing materials

Correct answer: As sensors.

Article 3: Fiber computer allows apparel to run apps and "understand" the wearer

1. What have been developed by MIT researchers?

A. A flexible telescope mirror designed for clothing

B. A robot system that automatically cooks food

C. A standard laptop that can be sewn onto fabric

D. Autonomous programmable computer as an elastic fiber.

Correct answer: Autonomous programmable computer as an elastic fiber.

2. What can the fiber computer monitor?

A. The price of clothing in online shops

B. Health conditions and physical activity.

C. The temperature of distant stars

D. Only internet browsing history

Correct answer: Health conditions and physical activity.

3. Why could smart fabrics be advantageous compared to regular wearables?

A. They never need sensors, batteries, or communication systems

B. They work only when placed flat on a laboratory desk

C. They are in contact with larger area of the body close to vital organs.

D. They replace all medical expertise with automatic decisions

Correct answer: They are in contact with larger area of the body close to vital organs.

4. What elements does the fiber computer consist of?

A. Sensors, microcontroller, memory, Bluetooth modules, optical communication, and battery.

B. A telescope mirror, a rocket engine, and navigation cameras

C. A keyboard, a mouse, a display, and a desktop processor

D. Only cotton threads, plastic buttons, and decorative fibers

Correct answer: Sensors, microcontroller, memory, Bluetooth modules, optical communication, and battery.

5. How many fiber computers were embedded into the top and leggings in the experiment?

A. Twenty-five

B. Four.

C. Zero

D. One hundred

Correct answer: Four.

6. What did each independently programmable fiber computer do in the experiment?

A. It printed images directly onto the surface of the fabric

B. It controlled a space telescope during a test observation

C. It sent automatic messages to social media platforms

D. It ran machine-learning model to recognize exercises performed by the wearer.

Correct answer: It ran machine-learning model to recognize exercises performed by the wearer.

7. What happened when fiber computers communicated with each other?

A. Their collective accuracy increased up to 95 percent.

B. They stopped functioning and became ordinary threads

C. They erased the activity data stored in the clothing

D. They required a large external computer to restart

Correct answer: Their collective accuracy increased up to 95 percent.

8. What makes the fiber computer clothing practical?

A. It required a large external computer to operate

B. It could only be used once before being discarded

C. It was comfortable, machine washable, and nearly undetectable.

D. It was rigid enough to prevent movement during exercise

Correct answer: It was comfortable, machine washable, and nearly undetectable.

9. What real-world test is conducted for fiber computer garments?

A. A mission to land a spacecraft on Mars

B. A cooking competition using robotic clothing

C. A race between autonomous cars and wearable sensors

D. Month-long winter research mission to the Arctic.

Correct answer: Month-long winter research mission to the Arctic.

10. What is the overall idea behind fiber computers in clothing?

A. Computers should be removed from wearable technology

B. Every-day apparel could gather, analyze, store, and communicate health and activity information.

C. Clothes should remain decorative and avoid functional sensing

D. Machine learning cannot operate inside fabric-based systems

Correct answer: Every-day apparel could gather, analyze, store, and communicate health and activity information.